



Original Investigation | Oncology

Stage IV Breast Cancer Incidence and Survival, 2010-2021

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Abstract

IMPORTANCE Trends in the incidence of de novo stage IV breast cancer in the general population and in the percentage with stage IV disease remain underreported.

OBJECTIVES To evaluate the incidence, percentage, and survival of people with de novo stage IV breast cancer diagnosed from 2010 through 2021.

DESIGN, SETTING, AND PARTICIPANTS This population-based cohort study used data from the Surveillance, Epidemiology, and End Results program to identify all individuals diagnosed with de novo invasive breast cancer between January 1, 2010, and December 31, 2021. No exclusion criteria were applied. Data analyses were conducted from January 2024 to June 2025.

MAIN OUTCOMES AND MEASURES Age-adjusted incidence rates of breast cancer and annual percentage changes (APC) from 2010 through 2021 were calculated. Among the diagnosed breast cancers, the yearly percentage of stage IV disease was also calculated. Changes in overall survival (OS) were evaluated for patients with stage IV breast cancer using multivariable Cox proportional hazards regression models stratified by tumor subtype.

RESULTS Of 761 471 breast cancer diagnoses (median [IQR] age at diagnosis, 60 [50-70] years; 99.2% female), 43 934 (5.8%) were stage IV. Stage IV incidence increased from 9.5 (95% CI, 9.2-9.9) cases per 100 000 females in 2010 to 11.2 (95% CI, 10.9-11.6) cases per 100 000 females in 2021, an APC of 1.2% (95% CI, 0.8%-1.6%). The incidence of stages I to III disease also increased, from 163.0 (95% CI, 161.6-164.4) cases per 100 000 females in 2010 to 177.4 (95% CI, 176.1-178.8) cases per 100 000 females in 2021. Stage IV incidence increased across all tumor subtypes, with an APC of 2.0% (95% CI, 1.5%-2.6%) for hormone receptor (HR)-positive/*ERBB2*-negative, 1.6% (95% CI, 0.2%-2.9%) for HR-positive/*ERBB2*-positive, 1.3% (95% CI, -0.1%-2.7%) for HR-negative/*ERBB2*-positive, and 2.7% (95% CI, 1.4%-4%) for triple-negative disease. Stage IV incidence increased statistically significantly across all age groups and numerically across all racial groups. Among males, there was a statistically significant increase in stage IV incidence, with an APC of 3.7% (95% CI, 1.0%-6.5%). The percentage of stage IV diagnoses increased statistically significantly from 2010 through 2021 in each tumor subtype: 4.2% to 4.4% for HR-positive/*ERBB2*-negative, 7.5% to 9.1% for HR-positive/*ERBB2*-positive, 9.3% to 10.4% for HR-negative/*ERBB2*-positive disease, and 6.1% to 7.6% for triple-negative breast cancer. For each successive year, OS among patients diagnosed with stage IV improved, with adjusted hazard ratios of 0.99 (95% CI, 0.98-0.99) for HR-positive/*ERBB2*-negative, 0.97 (95% CI, 0.95-0.99) for HR-positive/*ERBB2*-positive, and 0.97 (95% CI, 0.94-0.99) for HR-negative/*ERBB2*-positive; changes for triple-negative breast cancer were not statistically significant (AHR, 0.99 [95% CI, 0.97-1.01]; $P = .33$).

(continued)

Key Points

Question What were the changes in the incidence, percentage, and survival of individuals with de novo stage IV breast cancer diagnosed from 2010 through 2021?

Findings In this cohort study of 761 471 individuals with breast cancer, the incidence of stage IV breast cancer increased significantly by 1.2% per year, and the percentage of people with stage IV also increased significantly. Stage IV incidence increased widely across all ages, races, sexes, and tumor subtypes, while survival improved significantly from 2010 through 2021.

Meaning These findings suggest that efforts are needed to determine factors contributing to these increases and to identify breast cancer before patients present with de novo stage IV disease.

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Abstract (continued)

CONCLUSIONS AND RELEVANCE In this cohort study of individuals with stage IV breast cancer, incidence increased significantly overall, across ages, and for both sexes from 2010 through 2021. The percentage of individuals with stage IV vs stages I to III diagnoses also increased. Although OS improved, research is warranted to determine factors contributing to increased incidence, including potential changes in natural history of breast cancer, disease screening, and incidence and mortality of other conditions.

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Introduction

Breast cancer is the most common cancer diagnosed in females worldwide. Approximately 30% of patients develop metastases during the disease course.^{1,2} Breast cancer incidence rates have risen in the past 4 decades; from 2012 through 2021, the rate increased by 1% annually.³

Trends in the incidence of de novo stage IV breast cancer in the general population and trends in the percentage with stage IV among individuals diagnosed with breast cancer remain underreported. A previous study evaluating trends in incidence of distant disease in the US before 2010 showed a statistically significant increase in incidence for younger patients and a statistically significant decrease in older patients.⁴ A meta-analysis reported a decreasing percentage of stage IV presentation over time.⁵

More recent population-based work has highlighted heterogeneity in metastatic presentation and outcomes by age and tumor biology and has emphasized the need for contemporary incidence and survival estimates across stage categories.^{1,6} In addition, stage at diagnosis and metastatic outcomes differ by race and ethnicity and are shaped by differences in access to screening and timely diagnosis, structural inequities, and tumor subtype distribution. Collectively, these data underscore a persistent knowledge gap: contemporary US population-level estimates of time trends in de novo stage IV incidence, percentage with stage IV at diagnosis, and survival across age, race and ethnicity, tumor subtype, and sex, including males, remain limited.

We evaluated the time trends in age-adjusted incidence rates, percentage of stages I to III vs IV disease at presentation, and survival of patients diagnosed with de novo stage IV breast cancer in the US from 2010 through 2021. We hypothesized that the incidence and percentage of de novo stage IV breast cancer increased over time and that survival improved over the study period, with heterogeneity by tumor subtype and demographic subgroup. Considering the current increasing pattern in breast cancer incidence in the US,³ knowing whether this increase reflects increase in stages I to III or stage IV diagnoses, or both, has important health care policy and clinical implications.

Methods

Data Source

This was a retrospective, population-based cohort study of individuals diagnosed with invasive breast cancer from 2010 through 2021. Using data from the Surveillance, Epidemiology, and End Results (SEER) program 17 registry database, we included all individuals diagnosed with invasive breast cancer between January 1, 2010, and December 31, 2021; in situ (stage 0) cases were not included. No exclusion criteria were applied. Cases without histologic confirmation and cases identified by autopsy only or death certificate only were retained to preserve population-based incidence estimates. The SEER database includes information on cancer incidence, treatment, and survival for approximately 46% of the US population.⁷ The SEER program registries also collect data on patient demographics, primary tumor site, tumor morphology and stage at diagnosis, first course

of treatment, and follow-up for vital status. Although SEER is designed to capture geographically and demographically diverse populations, it is not a random national sample and may overrepresent certain regions and demographic groups. Given that we used deidentified data, this study was exempt from institutional review board. Because the study was not considered human participants' research, informed consent was waived. This study followed the Strengthening the Reporting of Observation Studies in Epidemiology (STROBE) reporting guideline for cohort studies.

Outcomes and Independent Variable for Temporal Analyses

The primary outcome was age-adjusted incidence rates of invasive breast cancer by stage and tumor subtype. We evaluated the percentage of stage IV disease among breast cancer diagnoses by calendar year. For patients with stage IV disease, overall survival (OS) and breast cancer-specific survival (BCSS) were evaluated. Calendar year of diagnosis was the primary independent variable for temporal analyses.

Covariates and Variable Classification

Race and ethnicity were obtained from SEER, as abstracted by tumor registrars. Race and ethnicity were assessed to describe the study population and to evaluate incidence. Categories included Hispanic (all races), non-Hispanic American Indian or Alaska Native, non-Hispanic Asian or Pacific Islander, non-Hispanic Black, non-Hispanic White, and unknown and were analyzed as reported in SEER. The variables analyzed from SEER for this study included age at diagnosis, race and ethnicity, sex, marital status, median household income in the county of residence at diagnosis, population size in area of residence, year of diagnosis, tumor grade, tumor histologic findings, tumor subtype, stage at diagnosis, type of surgery, chemotherapy, radiation therapy, sites of metastases, vital status, and cause of death.

Tumor subtype was classified into 5 subcategories based on the breast cancer subtype variable on SEER without modification: hormone receptor (HR)-positive/*ERBB2* (formerly *HER2*)-negative, HR-positive/*ERBB2*-positive, HR-negative/*ERBB2*-positive, triple-negative, and unknown. Invasive breast cancer cases were categorized based on SEER-derived American Joint Committee on Cancer, seventh edition, stage at initial diagnosis as stages I to III (ie, patients diagnosed with stage I, II, or III), stage IV (ie, patients diagnosed with de novo stage IV), and unknown when stage information was unavailable.

Statistical Analysis

Patient and tumor characteristics were calculated by frequency and percentage, and their distributions were compared between patients diagnosed with stages I to III vs stage IV breast cancer. Patients with unknown disease stage were not included in the comparisons. Comparisons used the χ^2 test. We calculated age-adjusted incidence rates by stage, tumor subtype, age, race and ethnicity, and sex. Age categories (<40, 40-49, 50-74, and >74 years) were defined a priori to reflect breast cancer screening recommendations during the study period rather than insurance eligibility thresholds. The incidence was reported as the number of new diagnoses in a given year per 100 000 population, age-adjusted to the 2000 US standard population using the 19 age groups available in SEER*Stat, version 8.4.2. We assessed annual percentage change (APC) in age-adjusted incidence rates for 2010 through 2021. The APC was calculated as $100 \times (\exp[\beta] - 1)$, with the regression coefficient (β) estimated by fitting the least-squares regression line to the natural logarithm of the rates, using calendar year as a regressor variable. The 95% CIs are provided for each APC, and we tested the hypothesis that APC equals 0.

Patients with missing data for a given variable were labeled as unknown for that variable and were included in all analyses. Variables with unknown categories are shown in the text, tables, and figures. Because missingness in registry-based variables may be nonrandom and unknown can represent meaningful clinical or documentation patterns, we retained these categories rather than performing multiple imputation.

We described the percentage of stage IV disease for each tumor subtype and evaluated trends in the percentage of stage IV breast cancer over time using the Cuzick nonparametric test for trend,⁸ because percentages reflect the distribution of stage at diagnosis among breast cancer cases rather than population-based incidence rates, for which APC methods are designed. As a descriptive analysis, we evaluated OS of patients with stage IV of each tumor subtype by using the Kaplan-Meier method without adjustments or formal log-rank comparisons. OS was defined as time from breast cancer diagnosis until death or last follow-up for surviving patients.

Among patients with stage IV breast cancer, we generated multivariable Cox proportional hazards regression models to assess changes in OS over time, with calendar year of diagnosis modeled as a continuous variable, stratified by tumor subtype and adjusted for age at diagnosis, race and ethnicity, sex, tumor grade, histologic findings, surgery, radiation therapy, chemotherapy, sites of distant metastases, marital status, median household income, rurality, and US region. These socioeconomic and geographic variables were included as covariates in survival analyses but were not incorporated into incidence APC models, which were population-based and descriptive in nature. We evaluated BCSS as a secondary end point in the same manner as OS. All *P* values reported were 2-sided, and *P* < .05 was considered statistically significant. Analyses were performed using Stata, version 12.0 (StataCorp LLC) and SPSS, version 26 (SPSS Inc). Data analysis was conducted from January 2024 to June 2025.

Results

Patient Characteristics

Among 761 471 individuals (median [IQR or range] age at diagnosis, 60 [50-70] years; 99.2% female and 0.8% male) diagnosed with invasive breast cancer from 2010 through 2021, 714 112 (93.8%) had stages I to III and 43 934 (5.8%) had stage IV disease at diagnosis. Individuals with stage IV breast cancer had a median (IQR) age of 63 (53-73) years, with 45.7% (*n* = 20 073) being 65 years or older and 18.4% (*n* = 8093) younger than 50 years of age. The race and ethnicity of the entire cohort included 12.4% Hispanic (all races), 0.6% non-Hispanic American Indian or Alaska Native, 9.4% non-Hispanic Asian or Pacific Islander, 10.5% non-Hispanic Black, and 66.4% non-Hispanic White individuals. In the analytic cohort, 8359 cases (1.09%) lacked histologic confirmation, 98 were autopsy-only, and 3043 were death certificate-only.

Compared with individuals with stages I-III disease, individuals with stage IV disease were more often non-Hispanic Black (15.3% vs 10.2%), had a lower median annual household income (eg, 25.1% vs 27.0% for \$90 000-\$119 999), and were less often married (42.3% vs 54.3%), less likely to have grade I tumors (4.3% vs 21.5%), and less likely to be HR-positive/*ERBB2*-negative (52.6% vs 70.3%) (**Table 1**). Other demographic and clinical characteristics of the overall study cohort are shown in Table 1 and are shown stratified by sex in eTable 1 in [Supplement 1](#).

Incidence of Breast Cancer From 2010 Through 2021

The age-adjusted incidence rate of stage IV breast cancer increased significantly from 9.5 (95% CI, 9.2-9.9) cases per 100 000 females in 2010 to 11.2 (95% CI, 10.9-11.6) cases per 100 000 females in 2021, an APC of 1.2% (95% CI, 0.8%-1.6%) (**Table 2**). Similarly, the age-adjusted incidence of stages I to III increased numerically from 163.0 (95% CI, 161.6-164.4) cases per 100 000 females in 2010 to 177.4 (95% CI, 176.1-178.8) cases per 100 000 females in 2021, an APC of 0.3% (95% CI, -0.3% to 0.9%), although this change was not statistically significant. The age-adjusted incidence of unknown stage also increased from 0.2 (95% CI, 0.2-0.3) cases per 100 000 females in 2010 to 0.6 (95% CI, 0.5-0.7) cases per 100 000 females in 2021, an APC of 11.3% (95% CI, 1.7%-21.9%). In sensitivity analyses excluding the COVID-19 period (2010-2019), the increase in stage IV incidence among females remained, with an APC of 1.0% (95% CI, 0.5%-1.5%), consistent with the primary analysis through 2021.

Table 1. Characteristics of Patients in SEER Diagnosed With Invasive Breast Cancer From 2010 Through 2021, by Stage at Diagnosis

Characteristic	Patients, No. (%)				P value ^a
	Stage I-III	Stage IV	Unknown stage	Total	
All patients	714 112 (93.8)	43 934 (5.8)	3425 (0.4)	761 471 (100.0)	NA
Age at diagnosis, y					
<50	136 203 (19.1)	8093 (18.4)	510 (14.9)	144 806 (19.0)	<.001
50-64	257 831 (36.1)	15 768 (35.9)	1013 (29.6)	274 612 (36.1)	
>64	320 078 (44.8)	20 073 (45.7)	1902 (55.5)	342 053 (44.9)	
Sex					
Female	708 753 (99.2)	43 400 (98.8)	3377 (98.6)	755 530 (99.2)	<.001
Male	5359 (0.8)	534 (1.2)	48 (1.4)	5941 (0.8)	
Race and ethnicity					
Hispanic (all races)	88 754 (12.4)	5454 (12.4)	481 (14.0)	94 689 (12.4)	<.001
Non-Hispanic American Indian or Alaska Native	4302 (0.6)	300 (0.7)	28 (0.8)	4630 (0.6)	
Non-Hispanic Asian or Pacific Islander	67 727 (9.5)	3562 (8.1)	272 (7.9)	71 561 (9.4)	
Non-Hispanic Black	72 794 (10.2)	6702 (15.3)	493 (14.4)	79 989 (10.5)	
Non-Hispanic White	476 173 (66.7)	27 760 (63.2)	2005 (58.5)	505 938 (66.4)	
Unknown	4362 (0.6)	156 (0.4)	146 (4.3)	4664 (0.6)	
Marital status at diagnosis					
Married	387 565 (54.3)	18 590 (42.3)	1070 (31.2)	407 225 (53.5)	<.001
Single	105 348 (14.8)	9664 (22.0)	509 (14.9)	115 521 (15.2)	
Domestic partner	3335 (0.5)	205 (0.5)	5 (0.1)	3545 (0.5)	
Other (separated, divorced, or widowed)	179 276 (25.1)	13 077 (29.8)	1041 (30.4)	193 394 (25.4)	
Unknown	38 588 (5.4)	2398 (5.5)	800 (23.4)	41 786 (5.5)	
Median household income, \$					
≥120 000	37 819 (5.3)	2022 (4.6)	140 (4.1)	39 981 (5.3)	<.001
90 000-119 999	192 962 (27.0)	11 014 (25.1)	759 (22.2)	204 735 (26.9)	
70 000-89 999	268 236 (37.6)	16 659 (37.9)	1520 (44.4)	286 415 (37.6)	
55 000-69 999	145 254 (20.3)	9123 (20.8)	596 (17.4)	154 973 (20.4)	
40 000-54 999	60 933 (8.5)	4417 (10.1)	378 (11.0)	65 728 (8.6)	
<40 000	8815 (1.2)	696 (1.6)	31 (0.9)	9542 (1.3)	
Unknown	93 (0.0)	3 (0.0)	1 (0.0)	97 (0.0)	
Rural-urban					
Metropolitan areas ≥1 million population	436 500 (61.1)	26 554 (60.4)	2121 (61.9)	465 175 (61.1)	<.001
Metropolitan areas of 250 000-1 million population	153 375 (21.5)	9272 (21.1)	673 (19.6)	163 320 (21.4)	
Metropolitan areas of <250 000 population	50 909 (7.1)	3184 (7.2)	295 (8.6)	54 388 (7.1)	
Nonmetropolitan adjacent to a metropolitan area	42 235 (5.9)	2789 (6.3)	184 (5.4)	45 208 (5.9)	
Nonmetropolitan not adjacent to a metropolitan area	30 201 (4.2)	2071 (4.7)	149 (4.4)	32 421 (4.3)	
Unknown	892 (0.1)	64 (0.1)	3 (0.1)	959 (0.1)	
Histologic findings					
Ductal	565 580 (79.2)	32 137 (73.1)	1693 (49.4)	599 410 (78.7)	<.001
Lobular	71 525 (10.0)	4593 (10.5)	198 (5.8)	76 316 (10.0)	
Ductal and lobular	35 400 (5.0)	1525 (3.5)	79 (2.3)	37 004 (4.9)	
Other	41 607 (5.8)	5679 (12.9)	1455 (42.5)	48 741 (6.4)	
Tumor grade					
I	153 794 (21.5)	1889 (4.3)	379 (11.1)	156 062 (20.5)	<.001
II	284 173 (39.8)	9427 (21.5)	804 (23.5)	294 404 (38.7)	
III/IV	170 168 (23.8)	10 304 (23.5)	587 (17.1)	181 059 (23.8)	
Unknown	105 977 (14.8)	22 314 (50.8)	1655 (48.3)	129 946 (17.1)	

(continued)

Table 1. Characteristics of Patients in SEER Diagnosed With Invasive Breast Cancer From 2010 Through 2021, by Stage at Diagnosis (continued)

Characteristic	Patients, No. (%)				P value ^a
	Stage I-III	Stage IV	Unknown stage	Total	
Tumor subtype					
HR+/ERBB2-	502 065 (70.3)	23 131 (52.6)	1059 (30.9)	526 255 (69.1)	<.001
HR+/ERBB2+	67 096 (9.4)	5906 (13.4)	217 (6.3)	73 219 (9.6)	
HR-/ERBB2+	27 271 (3.8)	3160 (7.2)	82 (2.4)	30 513 (4.0)	
Triple-negative	71 116 (10.0)	5138 (11.7)	278 (8.1)	76 532 (10.1)	
Unknown	46 564 (6.5)	6599 (15.0)	1789 (52.2)	54 952 (7.2)	

Abbreviations: HR, hormone receptor; NA, not applicable; SEER, Surveillance, Epidemiology, and End Results; +, positive; -, negative.

^a P value for χ^2 test comparing the distributions of characteristics between breast cancer stages I to III vs IV.

Table 2. Age-Adjusted Incidence of Invasive Breast Cancer Diagnosed in Females From 2010 Through 2021, by Stage at Diagnosis

Year of diagnosis	Breast cancer stage					
	IV (n = 43 400)		I-III (n = 708 753)		Unknown (n = 3377)	
	Age-adjusted incidence rate (95% CI) ^a	SE	Age-adjusted incidence rate (95% CI) ^a	SE	Age-adjusted incidence rate (95% CI) ^a	SE
2010	9.5 (9.2 to 9.9)	0.2	163.0 (161.6 to 164.4)	0.7	0.2 (0.2 to 0.3)	0.0
2011	9.7 (9.4 to 10.0)	0.2	167.1 (165.7 to 168.5)	0.7	0.2 (0.2 to 0.3)	0.0
2012	9.7 (9.3 to 10.0)	0.2	166.7 (165.3 to 168.1)	0.7	0.2 (0.1 to 0.2)	0.0
2013	10.2 (9.9 to 10.6)	0.2	167.8 (166.4 to 169.2)	0.7	0.2 (0.2 to 0.3)	0.0
2014	10.2 (9.9 to 10.6)	0.2	167.8 (166.5 to 169.2)	0.7	0.2 (0.1 to 0.2)	0.0
2015	9.9 (9.5 to 10.2)	0.2	170.8 (169.4 to 172.2)	0.7	0.2 (0.1 to 0.2)	0.0
2016	10.3 (9.9 to 10.6)	0.2	165.2 (163.9 to 166.6)	0.7	0.8 (0.7 to 0.9)	0.0
2017	10.1 (9.8 to 10.5)	0.2	165.7 (164.3 to 167.0)	0.7	0.7 (0.6 to 0.8)	0.0
2018	10.4 (10.1 to 10.8)	0.2	171.6 (170.2 to 173.0)	0.7	0.6 (0.6 to 0.7)	0.0
2019	10.6 (10.2 to 10.9)	0.2	175.3 (173.9 to 176.6)	0.7	0.5 (0.4 to 0.6)	0.0
2020	10.5 (10.2 to 10.9)	0.2	157.8 (156.5 to 159.2)	0.7	0.6 (0.5 to 0.6)	0.0
2021	11.2 (10.9 to 11.6)	0.2	177.4 (176.1 to 178.8)	0.7	0.6 (0.5 to 0.7)	0.0
2010 to 2021 APC ^b	1.2 (0.8 to 1.6)	NA	0.3 (-0.3 to 0.9)	NA	11.3 (1.7 to 21.9)	0.0

Abbreviations: APC, annual percentage change; NA, not available.

^b For APC estimates, a separate SE is not provided in the SEER*Stat exported output.

^a Age-adjusted incidence rates are presented per 100 000 persons with 95% CIs and SEs, as provided by SEER*Stat.

Incidence of Stage IV Breast Cancer From 2010 Through 2021 in Key Subgroups

Further analysis of the age-adjusted incidence rate of stage IV incidence showed statistically significant increases for triple-negative breast cancer, with an APC of 2.7% (95% CI, 1.4%-4.0%); HR-positive/ERBB2-negative, with an APC of 2.0% (95% CI, 1.5%-2.6%); and HR-positive/ERBB2-positive, with an APC of 1.6% (95% CI, 0.2%-2.9%). For HR-negative/ERBB2-positive disease, the APC was 1.3% (95% CI, -0.1% to 2.7%), which was not statistically significant (Table 3).

Table 4 shows the APC in incidence rates divided by age group and race and ethnicity for each tumor subtype. We observed a generalized increase in the age-adjusted incidence rates of stage IV breast cancer. Among all age groups, individuals younger than 40 years had the highest increase in overall incidence (APC, 3.1% [95% CI, 2.2%-4.0%]), which was driven by HR-positive cancers (Table 4). By contrast, individuals older than 74 years had the highest increase in triple-negative cancers (APC, 4.3% [95% CI, 1.9%-6.8%]). The age-adjusted incidence of stage IV breast cancer increased numerically across all racial and ethnic groups, with the highest overall magnitude in non-Hispanic Asian and Pacific Islander individuals (APC, 3.4% [95% CI, 2.4%-4.4%]).

In males, we observed a significant increase in age-adjusted incidence rates. Rates increased from 0.12 (95% CI, 0.09-0.17) cases per 100 000 males in 2010 to 0.20 (95% CI, 0.16-0.26) cases per 100 000 males in 2021, an APC of 3.7% (95% CI, 1.0%-6.5%) (eTable 2 in Supplement 1).

Table 3. Age-Adjusted Incidence of Stage IV Breast Cancer Diagnosed in Females From 2010 Through 2021, Overall and by Tumor Subtype

Year of diagnosis	Overall (n = 43 400)		HR+/ERBB2- (n = 22 798)		HR+/ERBB2+ (n = 5833)		HR-/ERBB2+ (n = 3152)		Triple-negative (n = 5106)		Unknown subtype (n = 6511)	
	Age-adjusted incidence rate (95% CI) ^a	SE	Age-adjusted incidence rate (95% CI) ^a	SE	Age-adjusted incidence rate (95% CI) ^a	SE	Age-adjusted incidence rate (95% CI) ^a	SE	Age-adjusted incidence rate (95% CI) ^a	SE	Age-adjusted incidence rate (95% CI) ^a	SE
2010	9.5 (9.2 to 9.9)	0.2	4.6 (4.4 to 4.9)	0.1	1.2 (1.1 to 1.3)	0.1	0.7 (0.6 to 0.7)	0.0	1.1 (1.0 to 1.3)	0.1	1.9 (1.8 to 2.1)	2.1
2011	9.7 (9.4 to 10.0)	0.2	5 (4.8 to 5.2)	0.1	1.2 (1.1 to 1.4)	0.1	0.7 (0.6 to 0.8)	0.0	1.1 (1.0 to 1.2)	0.1	1.7 (1.5 to 1.8)	1.8
2012	9.7 (9.3 to 10.0)	0.2	5 (4.7 to 5.2)	0.1	1.4 (1.2 to 1.5)	0.1	0.7 (0.6 to 0.8)	0.0	1.1 (1.0 to 1.2)	0.1	1.6 (1.5 to 1.7)	1.7
2013	10.2 (9.9 to 10.6)	0.2	5.2 (5.0 to 5.4)	0.1	1.4 (1.3 to 1.5)	0.1	0.8 (0.7 to 0.9)	0.1	1.2 (1.0 to 1.3)	0.1	1.7 (1.5 to 1.8)	1.8
2014	10.2 (9.9 to 10.6)	0.2	5.2 (5.0 to 5.5)	0.1	1.4 (1.3 to 1.6)	0.1	0.8 (0.7 to 0.9)	0.0	1.1 (1.0 to 1.3)	0.1	1.6 (1.5 to 1.8)	1.8
2015	9.9 (9.5 to 10.2)	0.2	5 (4.7 to 5.2)	0.1	1.5 (1.4 to 1.7)	0.1	0.8 (0.7 to 0.9)	0.0	1.1 (1.0 to 1.2)	0.1	1.5 (1.4 to 1.7)	1.7
2016	10.3 (9.9 to 10.6)	0.2	5.2 (5.0 to 5.5)	0.1	1.6 (1.4 to 1.7)	0.1	0.8 (0.7 to 0.9)	0.0	1.2 (1.1 to 1.3)	0.1	1.5 (1.4 to 1.6)	1.6
2017	10.1 (9.8 to 10.5)	0.2	5.4 (5.1 to 5.6)	0.1	1.5 (1.3 to 1.6)	0.1	0.8 (0.7 to 0.9)	0.0	1.1 (1.0 to 1.2)	0.1	1.3 (1.2 to 1.5)	1.5
2018	10.4 (10.1 to 10.8)	0.2	5.8 (5.6 to 6.1)	0.1	1.4 (1.3 to 1.5)	0.1	0.7 (0.6 to 0.8)	0.0	1.3 (1.2 to 1.5)	0.1	1.2 (1.1 to 1.3)	1.3
2019	10.6 (10.2 to 10.9)	0.2	5.8 (5.5 to 6.0)	0.1	1.4 (1.3 to 1.6)	0.1	0.8 (0.7 to 0.9)	0.0	1.4 (1.2 to 1.5)	0.1	1.2 (1.1 to 1.4)	1.4
2020	10.5 (10.2 to 10.9)	0.2	5.8 (5.5 to 6.0)	0.1	1.4 (1.2 to 1.5)	0.1	0.8 (0.7 to 0.9)	0.0	1.3 (1.2 to 1.4)	0.1	1.3 (1.2 to 1.4)	1.4
2021	11.2 (10.9 to 11.6)	0.2	6 (5.7 to 6.2)	0.1	1.6 (1.4 to 1.7)	0.1	0.8 (0.7 to 0.9)	0.0	1.5 (1.4 to 1.7)	0.1	1.4 (1.3 to 1.5)	1.5
2010 to 2021 APC ^b	1.2 (0.8 to 1.6)	NA	2.0 (1.5 to 2.6)	NA	1.6 (0.2 to 2.9)	NA	1.3 (-0.1 to 2.7)	NA	2.7 (1.4 to 4.0)	NA	-3.4 (-4.6 to -2.1)	NA

Abbreviations: APC, annual percentage change; HR, hormone receptor; NA, not applicable; +, positive; -, negative.

^a Age-adjusted incidence rates are presented per 100 000 persons with 95% CIs and SEs, as provided by SEER*Stat.

^b For APC estimates, a separate SE is not provided in the SEER*Stat exported output.

Percentages of Stage IV Breast Cancer Among Existing Breast Cancer Cases From 2010 Through 2021

Across the study period (2010-2021), the percentage of individuals with stage IV breast cancer at diagnosis was 5.8% (eTable 3 in Supplement 1). The percentages of stages I to III vs stage IV breast cancer varied significantly by tumor subtype: 95.4% vs 4.4% in HR-positive/*ERBB2*-negative tumors, 91.6% vs 8.1% in HR-positive/*ERBB2*-positive breast cancer, 89.4% vs 10.4% in HR-negative/*ERBB2*-positive disease, and 92.9% vs 6.7%, respectively, for individuals with triple-negative breast cancer (overall $P < .001$).

The percentages of stage IV disease increased modestly from 5.6% in 2010 to 6.0% in 2021 ($P < .001$ for trend). Increases were observed across tumor subtypes, including HR-positive/*ERBB2*-negative (4.2% to 4.4%), HR-positive/*ERBB2*-positive (7.5% to 9.1%), HR-negative/*ERBB2*-positive (9.3% to 10.4%), and triple-negative disease (6.1% to 7.6%) (Figure; eTable 4 in Supplement 1). In addition, cases classified as unknown increased over time, reaching 14.4% by 2019 and 16.9% by 2021.

Survival Analyses for Patients With Stage IV Breast Cancer

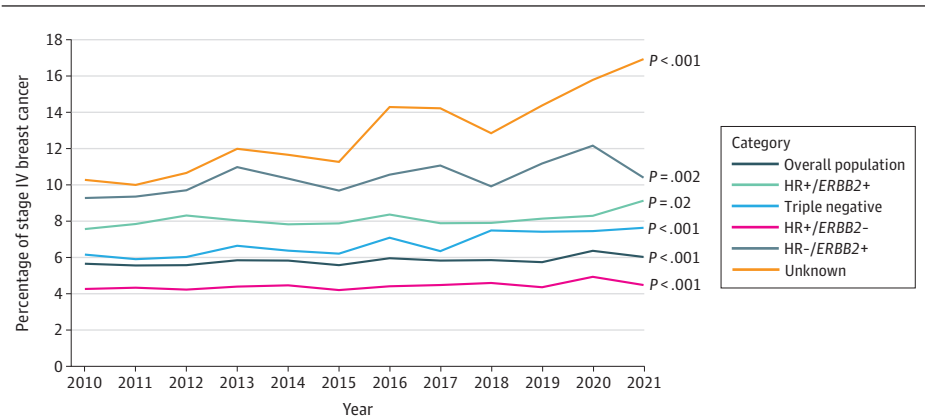
Patients diagnosed with stage IV breast cancer ($n = 43\,934$) had a median (IQR) follow-up of 63 (31-99) months. Over this period, there were 28 888 deaths in this group, of which 25 317 (87.6%)

Table 4. Annual Percentage Change in Age-Adjusted Incidence of Stage IV Breast Cancer in Females, by Age at Diagnosis and by Race and Ethnicity

Characteristic	No.	Annual percentage change (95% CI)					
		Overall (n = 43 934)	HR+/ <i>ERBB2</i> - (n = 23 131)	HR+/ <i>ERBB2</i> + (n = 5906)	HR-/ <i>ERBB2</i> + (n = 3160)	Triple-negative (n = 5138)	Unknown (n = 6599)
Overall		1.2 (0.8 to 1.6)	2 (1.5 to 2.6)	1.6 (0.2 to 2.9)	1.3 (-0.1 to 2.7)	2.7 (1.4 to 4.0)	-3.4 (-4.6 to -2.1)
Age at diagnosis, y							
<40	2783	3.1 (2.2 to 4.0)	4.5 (2.9 to 6.1)	3.4 (0.9 to 5.9)	0.8 (-4.1 to 6.0)	1.8 (-2.0 to 5.6)	-1.3 (-6.1 to 3.7)
40 to 49	5310	1.7 (0.5 to 2.9)	2.0 (0.5 to 3.6)	2.7 (0.9 to 4.5)	1.6 (-0.7 to 4.0)	2.5 (-1.1 to 6.2)	-2.4 (-5.3 to 0.5)
50 to 74	25 953	0.5 (0.1 to 0.9)	1.2 (0.4 to 2.0)	0.4 (-1.0 to 1.8)	1.2 (0.2 to 2.3)	2.2 (1.2 to 3.3)	-3.9 (-5.4 to -2.3)
>74	9888	1.7 (0.4 to 3)	3.3 (1.8 to 4.8)	2.1 (-0.6 to 4.9)	1.1 (-1.4 to 3.7)	4.3 (1.9 to 6.8)	-3.0 (-5.2 to -0.8)
Race and ethnicity							
Hispanic	5454	1.9 (1.0 to 2.9)	2.6 (1.5 to 3.8)	1.3 (-1.3 to 3.9)	2.5 (-0.9 to 6.0)	5.3 (2.7 to 8.0)	-2.5 (-4.2 to -0.7)
Non-Hispanic American Indian or Alaska Native	300	1.9 (-2.0 to 5.9)	4.5 (-1.5 to 10.8)	0.6 (-9.2 to 11.5)	NA	NA	-4.8 (-14.2 to 5.8)
Non-Hispanic Asian or Pacific Islander	3562	3.4 (2.4 to 4.4)	3.6 (2.1 to 5.2)	5.7 (2.5 to 8.9)	2.6 (-1.3 to 6.6)	3.2 (-1.0 to 7.6)	-0.8 (-3.8 to 2.4)
Non-Hispanic Black	6702	0.5 (-0.2 to 1.3)	1.6 (0.3 to 2.9)	-1.1 (-2.6 to 0.5)	0.6 (-1.2 to 2.5)	2.3 (0.8 to 3.9)	-3.5 (-5.9 to -1.0)
Non-Hispanic White	27 760	1.0 (0.6 to 1.5)	2.0 (1.3 to 2.7)	1.8 (0.1 to 3.7)	0.6 (-0.9 to 2.1)	2.0 (0.7 to 3.3)	-3.7 (-5.0 to -2.3)

Abbreviations: HR, hormone receptor; NA, not applicable; +, positive; -, negative.

Figure. Line Graph Showing Percentages of Breast Cancers Diagnosed as Stage IV in Successive Years From 2010 Through 2021, Overall and by Tumor Subtype



HR indicates hormone receptor; +, positive; -, negative.

were due to breast cancer. The rate of OS at 5 years was 30.6% (95% CI, 29.9%-31.3%) for HR-positive/*ERBB2*-negative breast cancer, 41.3% (95% CI, 39.8%-42.7%) for HR-positive/*ERBB2*-positive disease, 35.5% (95% CI, 33.6%-37.5%) for HR-negative/*ERBB2*-positive tumors, and 10.3% (95% CI, 9.3%-11.4%) for patients with triple-negative breast cancer (eFigure 1 in [Supplement 1](#)).

In adjusted Cox models, OS improved significantly over time for each successive year among patients with HR-positive/*ERBB2*-negative disease (adjusted hazard ratio [AHR], 0.99 [95% CI, 0.98-0.99]; $P = .04$), for HR-positive/*ERBB2*-positive tumors (AHR, 0.97 [95% CI, 0.95-0.99]; $P = .003$), and for patients with HR-negative/*ERBB2*-positive breast cancer (AHR, 0.97 [95% CI, 0.94-0.99]; $P = .02$). The improvement in OS over time in triple-negative breast cancer was not statistically significant (AHR, 0.99 [95% CI, 0.97-1.01]; $P = .33$) (eTable 5 in [Supplement 1](#)). BCSS results were similar to those of OS both for the unadjusted survival rates by tumor subtype (eFigure 2 in [Supplement 1](#)) and for the adjusted Cox models (eTable 6 in [Supplement 1](#)).

Discussion

This study assessed the time trends in age-adjusted incidence rates, percentages with disease stages I to III vs IV at presentation, and survival of patients diagnosed with de novo stage IV breast cancer in the US from 2010 through 2021. Our study showed that the incidence of de novo stage IV breast cancer increased numerically year-over-year across all tumor subtypes, all age groups, both sexes, and most races and ethnicities. Population-level data describing temporal trends in de novo metastatic breast cancer among males are limited, and male breast cancer remains relatively understudied. The parallel increase in stage IV incidence in both sexes suggests that the observed trend may reflect broader environmental or population-level factors influencing breast cancer risk and stage at presentation, rather than mechanisms unique to female-specific exposures. We also observed a rising incidence of stages I-III and unknown stage disease, suggesting an overall increase in breast cancer diagnoses rather than simple stage reclassification. Prior analyses reported stable stage IV incidence before 2010 despite screening expansion.^{4,9,10} In contrast, our findings demonstrate a significant increase beginning in 2010, including among screening-eligible age groups. The underlying drivers remain uncertain and may reflect population-level changes in risk factors, screening patterns, or access to care.

Both incidence and percentage of stage IV disease increased. Notably, incidence of stages I to III increased while the percentage decreased, indicating that the increase in stage IV reflects both absolute growth and a shifting distribution at diagnosis. Percentages varied by subtype, ranging from 4.4% for HR-positive/*ERBB2*-negative to 10.4% for HR-negative/*ERBB2*-positive disease. These findings align with prior reports of increasing metastatic presentation in certain subgroups, including younger patients.^{5,11-13}

Consistent with existing literature,¹⁴⁻¹⁸ disparities in stage at diagnosis persisted. Non-Hispanic Black individuals and individuals with lower income were more likely to present with stage IV disease, supporting the continued importance of social determinants of health and access to care.

Survival improved over time for most subtypes, reflecting advances in systemic therapy.^{19,20} Improvements were not statistically significant for triple-negative disease, underscoring ongoing therapeutic challenges in this subgroup.

The increase in de novo stage IV incidence has important clinical and public health implications. Efforts to ensure equitable access to timely screening and diagnostic evaluation remain critical, particularly in populations with rising incidence and established disparities. The consistent increase across subtypes and in both females and males also highlights the need to investigate broader population-level risk factors and develop targeted strategies to reduce late-stage presentation.

Strengths and Limitations

This study has several strengths. These include the large, population-based design and the simultaneous evaluation of incidence, stage distribution, and survival across demographic and tumor subgroups.

Our study has some limitations. First, it contains data from the US, and incidence and outcome trends may differ in other countries. However, breast cancer incidence is increasing in most countries worldwide.^{21,22} Second, SEER does not have information about breast cancer screening use or adherence, which would complement the interpretation of the results in certain age groups. Third, SEER does not have information about epidemiological risk factors for breast cancer, such as obesity, alcohol use, breast feeding, and hormone use. While the rates of obesity have increased in the US over the years, the interplay with other factors cannot be determined in our study. The absence of these variables limits our ability to assess how differences in screening practices, comorbidity burden, and individual-level socioeconomic or lifestyle factors may have contributed to observed incidence and stage-at-diagnosis trends. Although SEER captures area-level socioeconomic indicators and geographic region, our primary incidence analyses were designed as descriptive population-based trend assessments rather than etiologic models of determinants of late-stage presentation. Accordingly, incidence APC estimates were not stratified or adjusted by region or socioeconomic status. We acknowledge that temporal trends in stage IV incidence may vary across geographic regions or health policy environments, including differences in Medicaid expansion or access to care, and this represents an important direction for future investigation.

Fourth, we evaluated patients with de novo stage IV breast cancer, and the outcomes for patients with recurrent metastatic disease (which is not captured in SEER) may be different. Fifth, cases identified by death certificate or autopsy and cases without histologic confirmation may have less complete information on stage and tumor biomarkers, which could contribute to unknown categories in subgroup analyses. Sixth, given the number of statistical comparisons performed, the use of a nominal 2-sided $P < .05$ threshold may have resulted in some false-positive findings, and results should be interpreted in this context. Lastly, health care access and cancer detection were disrupted during the COVID-19 pandemic. Reductions in routine screening and delays in diagnostic evaluation in 2020 may have led to temporary underdiagnosis of breast cancer, followed by catch-up diagnoses in 2021. Such delays could plausibly contribute to a higher likelihood of advanced-stage presentation if tumors progressed during periods of reduced health care access. We also observed a sharp increase in cases classified as unknown subtype between 2019 and 2021, which may reflect disruptions in diagnostic workup, incomplete biomarker testing, or delays in pathology reporting during the pandemic period rather than true biologic shifts in tumor distribution. Importantly, sensitivity analyses restricted to 2010 to 2019 demonstrated similar increasing trends in stage IV incidence, suggesting that the overall findings are not solely attributable to pandemic-related effects.

Conclusions

In conclusion, this cohort study found that from 2010 to 2021 the incidence and percentage of de novo stage IV breast cancer diagnoses increased in the US, including among males, despite improvements in survival. These findings underscore the need to better understand drivers of advanced-stage cancer presentation and to develop strategies to reduce the burden of metastatic disease at diagnosis.

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REFERENCES

1. Mariotto AB, Etzioni R, Hurlbert M, Penberthy L, Mayer M. Estimation of the number of women living with metastatic breast cancer in the United States. *Cancer Epidemiol Biomarkers Prev*. 2017;26(6):809-815. doi:10.1158/1055-9965.EPI-16-0889
2. Łukasiewicz S, Czelelewski M, Forma A, Baj J, Sitarz R, Stanisławek A. Breast cancer-epidemiology, risk factors, classification, prognostic markers, and current treatment strategies—an updated review. *Cancers (Basel)*. 2021;13(17):4287. doi:10.3390/cancers13174287
3. Giaquinto AN, Sung H, Newman LA, et al. Breast cancer statistics 2024. *CA Cancer J Clin*. 2024;74(6):477-495.
4. Johnson RH, Chien FL, Bleyer A. Incidence of breast cancer with distant involvement among women in the United States, 1976 to 2009. *JAMA*. 2013;309(8):800-805. doi:10.1001/jama.2013.776
5. Benitez Fuentes JD, Morgan E, de Luna Aguilar A, et al. Global stage distribution of breast cancer at diagnosis: a systematic review and meta-analysis. *JAMA Oncol*. 2024;10(1):71-78. doi:10.1001/jamaoncol.2023.4837
6. Thomas A, Rhoads A, Pinkerton E, et al. Incidence and survival among young women with stage I-III breast cancer: SEER 2000-2015. *J Natl Cancer Inst Cancer Spectr*. 2019;3(3):pkz040. doi:10.1093/jncics/pkz040
7. Overview of the SEER Program. National Cancer Institute Surveillance, Epidemiology and End Results Program. Accessed March 27, 2026. <https://seer.cancer.gov/about/overview.html>
8. Cuzick J. A Wilcoxon-type test for trend. *Stat Med*. 1985;4(1):87-90. doi:10.1002/sim.4780040112
9. Bleyer A, Welch HG. Effect of three decades of screening mammography on breast-cancer incidence. *N Engl J Med*. 2012;367(21):1998-2005. doi:10.1056/NEJMoa1206809
10. Esserman L, Shieh Y, Thompson I. Rethinking screening for breast cancer and prostate cancer. *JAMA*. 2009;302(15):1685-1692. doi:10.1001/jama.2009.1498
11. Lannin DR, Wang S. Are small breast cancers good because they are small or small because they are good? *N Engl J Med*. 2017;376(23):2286-2291. doi:10.1056/NEJMs1613680
12. Yao K, Tomasz C, Pesce C, et al Abstract P3-06-14: Increase in the proportion of stage IV breast cancer amongst young women with newly diagnosed breast cancer, report from the national cancer data base, 1998-2009. *Cancer Res*. 2013;73(24_supplement):P3-06-14.
13. Nicholson WK, Silverstein M, Wong JB, et al; US Preventive Services Task Force. Screening for breast cancer: US Preventive Services Task Force recommendation statement. *JAMA*. 2024;331(22):1918-1930. doi:10.1001/jama.2024.5534
14. Williams F, Thompson E. Disparities in breast cancer stage at diagnosis: importance of race, poverty, and age. *J Health Dispar Res Pract*. 2017;10(3):34-45.
15. Chen L, Li CI. Racial disparities in breast cancer diagnosis and treatment by hormone receptor and HER2 status. *Cancer Epidemiol Biomarkers Prev*. 2015;24(11):1666-1672. doi:10.1158/1055-9965.EPI-15-0293
16. Parise CA, Caggiano V. Regional variation in disparities in breast cancer specific mortality due to race/ethnicity, socioeconomic status, and urbanization. *J Racial Ethn Health Disparities*. 2017;4(4):706-717. doi:10.1007/s40615-016-0274-4
17. Zhu S, Lei C. Association between marital status and all-cause mortality of patients with metastatic breast cancer: a population-based study. *Sci Rep*. 2023;13(1):9067. doi:10.1038/s41598-023-36139-8
18. Yuan R, Zhang C, Li Q, Ji M, He N. The impact of marital status on stage at diagnosis and survival of female patients with breast and gynecologic cancers: a meta-analysis. *Gynecol Oncol*. 2021;162(3):778-787. doi:10.1016/j.ygyno.2021.06.008
19. Welt A, Bogner S, Arendt M, et al. Improved survival in metastatic breast cancer: results from a 20-year study involving 1033 women treated at a single comprehensive cancer center. *J Cancer Res Clin Oncol*. 2020;146(6):1559-1566. doi:10.1007/s00432-020-03184-z
20. Giordano SH, Buzdar AU, Smith TL, Kau SW, Yang Y, Hortobagyi GN. Is breast cancer survival improving? *Cancer*. 2004;100(1):44-52. doi:10.1002/cncr.11859
21. Bray F, Ferlay J, Soerjomataram I, Siegel RL, Torre LA, Jemal A. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin*. 2018;68(6):394-424. doi:10.3322/caac.21492

22. van der Meer DJ, Kramer I, van Maaren MC, et al. Comprehensive trends in incidence, treatment, survival and mortality of first primary invasive breast cancer stratified by age, stage and receptor subtype in the Netherlands between 1989 and 2017. *Int J Cancer*. 2021;148(9):2289-2303. doi:10.1002/ijc.33417

SUPPLEMENT 1.

eTable 1. Characteristics of patients in SEER diagnosed with invasive breast cancer between 2010 and 2021, according to sex and stage at diagnosis

eTable 2. Age-adjusted Incidence of stage IV breast cancer diagnoses in men between 2010-2021 overall and by tumor subtype

eTable 3. Proportion of breast cancers diagnosed as stage IV breast cancer between 2010-2021 by tumor subtype

eTable 4. Proportions of breast cancers diagnosed as stage IV in successive years from 2010 to 2021, by tumor subtype

eTable 5. Stratified multivariable Cox models showing trends in overall survival over time in patients diagnosed with de novo stage IV breast cancer by tumor subtype

eTable 6. Stratified multivariable Cox models showing trends in breast cancer-specific survival (BCSS) over time in patients diagnosed with de novo stage IV breast cancer by tumor subtype

eFigure 1. Overall survival by tumor subtype among patients diagnosed with de novo stage IV disease between 2010-2021

eFigure 2. Breast cancer-specific survival by tumor subtype among patients diagnosed with de novo stage IV disease between 2010-2021

SUPPLEMENT 2.

Data Sharing Statement